



MATERIAL SAFETY DATA SHEET

Product	Metformin Hydrochloride	Issue Date	21 JUL 2021
MSDS No.	SHPL/MSDS/01/04	Effective Date	24 JUL 2021
Supersedes	SHPL/MSDS/01/03	Next Review Date	23 JUL 2024

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Trade Name : Metformin Hydrochloride
CAS Number : 1115-70-4
EC Number : 214-230-6

1.2 Relevant identified uses of the substance or mixture and uses advised against
No further relevant information available.

Application of the substance / the mixture

The substance is used as an Active Pharmaceutical Ingredient.

1.3 Details of the supplier of the safety data sheet

Manufacturer/Supplier:

Sohan Healthcare Pvt. Ltd.

D-30, Kurkumbh, MIDC,

Tal - Daund, District - Pune -413802 (Maharashtra) INDIA.

Further information obtainable from:

Tel: +91-02117-202033

Tel/Fax:91-2117-235333

E-Mail:sales@sohanhealthcare.com

1.4 Emergency telephone number:

Emergency telephone number: 09822278411

Opening hours: 09:00 AM to 05:30 PM

Other Comments (e.g. Language(s) of the phone service): English, Hindi, Marathi

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Designation	Executive	Manager	Manager
Department	EHS	Production	QA
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SECTION 2: Hazards identification

2.1 Classification of the substance/mixture

Classification according to Regulation (EC) No. 1272/2008



Acute Tox. 4 : H302 Harmful if swallowed
 Aquatic Chronic 3 : H412 Harmful to aquatic life with long lasting effects.

2.2 Label elements

Labeling according to Regulation (EC) No. 1272/2008

The substance is classified and labeled according to the CLP regulation.

Hazard pictograms



GHS07

Signal word Warning

Hazard statements

H302 Harmful if swallowed.
 H412 Harmful to aquatic life with long lasting effects.
 H319 causes serious eye irritation.

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Precautionary statements

P273 Avoid release to the environment.
P264 Wash thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P280 Wear eye protection/face protection.
P301+P312 IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P337+P313 If eye irritation persists: Get medical advice/attention.
P330 Rinse mouth.
P501 Dispose of contents/container in accordance with local/regional/national/International regulations.

2.3 Other hazards

Results of PBT and vPvB assessment

PBT : Not applicable.
vPvB : Not applicable.

SECTION 3: Composition/information on ingredients

3.1 Chemical characterisation: Substances

CAS No. 1115-70-4

Description : Metformin Hydrochloride

Identification number(s)

EC number : 214-230-6

RTECS : DU1800000

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Additional information:

Molecular formula: $C_4H_{11}N_5.HCl$

Molecular weight: 165.62 g/mol

Purity: 98.5% to 101.0%

SECTION 4: First aid measures

4.1 Description of first aid measures

General information:

Symptoms of poisoning may even occur after several hours; therefore, medical observation for at least 48 hours after the accident is recommended.

After inhalation: Supply fresh air; consult doctor in case of complaints.

After skin contact: Generally, the product does not irritate the skin. Immediately rinse with water.

After eye contact: Rinse opened eye for several minutes under running water. If symptoms persist, consult a doctor.

After swallowing: Rinse out mouth and then drink plenty of water. Call for a doctor immediately. Do not induce vomiting. Seek medical treatment.

4.2 Most important symptoms and effects, both acute and delayed: Diarrhea, nausea, vomiting

Information for doctor: Treat symptomatically and supportively.

4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available.

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SECTION 5: Firefighting measures**5.1 Extinguishing media**

Suitable extinguishing agents: Foam, Dry Chemical Powder, Carbon dioxide

5.2 Special hazards arising from the substance or mixture

In case of fire, the following can be released: Hydrogen Chloride (HCl)

5.3 Advice for fire fighters**Protective equipment:**

Wear self-contained breathing apparatus and protective clothing to prevent contact with skin.

Additional information

Prevent fire-extinguishing water from contaminating surface water or the ground water system. Suppress (knock down) gases/vapors/mists with a water spray jet.

SECTION 6: Accidental release measures**6.1 Personal precautions, protective equipment and emergency procedures**

Wear protective equipment. Keep unprotected persons away.

Ensure adequate ventilation. Avoid formation of dust.

6.2 Environmental precautions:

Do not allow product to reach sewage system or any water course.

Inform respective authorities in case of seepage into water course or sewage system.

6.3 Methods and material for containment and cleaning up:

Spills should be shoveled or vacuumed and placed into closed containers for disposal.

6.4 Reference to other sections

See Section 7 for information on safe handling.

See Section 8 for information on personal protection equipment.

See Section 13 for disposal information.

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SECTION 7: Handling and storage

7.1 Precautions for safe handling: Store in cool, dry place in tightly closed receptacles.
Information about fire-and explosion protection: Normal measures for preventive fire protection.

7.2 Conditions for safe storage, including any incompatibilities

Storage:

Requirements to be met by store room and receptacles:

Store in well closed containers.

Information about storage in one common storage facility:

Store away from oxidizing agents.

Store away from strong bases.

Store away from foodstuffs.

Further information about storage conditions: Keep container tightly sealed.

7.3 Specific end use(s) The substance is used as an Active Pharmaceutical Ingredient.

SECTION 8: Exposure controls/personal protection

Additional information about design of technical facilities:

Technical measures and appropriate working operations should be given priority over the use of personal protective equipment.

8.1 Control parameters

Ingredients with limit values that require monitoring at the workplace: Not required.

Additional information: lists used were valid at the time of SDS preparation.

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8.2 Exposure controls

Personal protective equipment:

General protective and hygienic measures: Keep away from food stuffs, beverages and feed.

Wash hands before breaks and at the end of work.

Immediately remove all soiled and contaminated clothing.

Avoid contact with eyes & skin.

Respiratory protection:

Required when dusts are generated.

Recommended Filter type: Filter P2 (acc. to DIN3181) for solid and liquid particles of harmful substances

Protection of hands:



Protective gloves

The glove material has to be impermeable and resistant to the product/the substance /the preparation.

Due to missing tests no recommendation to the glove material can be given for the product/preparation /chemical mixture.

Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation

Material of gloves: Nitrile rubber gloves, NBR.

The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer.

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Penetration time of glove material: Splas hand full contact

Glove thickness: 0.11 mm Break through time: >480 min

Eye protection: Safety glasses

Body protection: Protective work clothing

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

General Information

Appearance:

Form: Solid

Colour: White

Odour: Odourless

pH-value: Not applicable.

Change in condition

Melting point/Melting range: 222-226°C

Boiling point/Boiling range: Undetermined.

Flashpoint: Not applicable.

Flammability (solid, gaseous): Product is not flammable.

Ignition temperature: >402°C

Self-igniting: Product is not self-igniting.

Danger of explosion: Product does not present an explosion hazard.

Explosion limits:

Oxidising properties: No oxidizing properties.

Vapour pressure at 25°C : 0.001Pa

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Density:

Relative density at 20°C : 1.36

Solubility in/Miscibility with water at 20°C: 350 g/l

Partition coefficient (n-octanol/water) at 20°C: -3.5 log POW

Viscosity:

Dynamic: Not applicable.

Kinematic: Not applicable.

9.2 Other information :

Surface tension: 73 mN/m at 20°C

SECTION 10: Stability and reactivity

10.1 Reactivity

Stable at ambient temperature and under normal conditions of use.

10.2 Chemical stability Stable under normal conditions.

thermal decomposition /conditions to be avoided:

No decomposition if used according to specifications.

Formation of toxic gases is possible during heating or in case of fire.

10.3 Possibility of hazardous reactions

Violent reactions possible with strong oxidizing agents and bases.

10.4 Conditions to avoid: Strong heating (decomposition).

10.5 Incompatible materials: Incompatible with oxidising agents and bases.

10.6 Hazardous decomposition products:

No dangerous decomposition products known.

Fire may cause evolution of Hydrogen chloride gas.

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SECTION 11: Toxicological information

11.1 Information on toxicological effects:

Acute toxicity: Harmful if swallowed.

LD/LC50 values relevant for classification:
375 mg/kg bw (dog)
552 mg/kg bw(rabbit)
1770 mg/kg (rat)

Primary irritant effect:

Skin corrosion / irritation

The test material is not irritant to the skin of rabbits in this study when tested according to the AFNOR protocol in comparison to the OECD protocol.

Serious eye damage/irritation

The Mean Ocular Irritation Index of Metformin HCl amounts to 4.17 after 4days (without rinsing). These effects are fully reversible after 7 days.

Respiratory or skin sensitisation

No sensitising effects known.

QSAR calculations demonstrate the absence of any potential for sensitisation. This is supported by experience from long term work place exposure with no reported evidence for skin sensitisation in humans.

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Repeated dose toxicity

Oral route:

The study was designed to characterize the potential chronic toxicity of a test material, Metformin Hydrochloride, when administered daily in the diet of Cr1:CD®-1 (ICR)BR mice for at least 52 consecutive weeks.

Based upon the results of this study, daily administration of Metformin HCl to mice for at least 52 consecutive weeks was well tolerated at dietary levels of 450 mg/kg/day; while a histomorphological teration in the kidneys was observed in mice following at least 52 weeks of dietary administration of 1500 mg/kg/day of Metformin HCl.

CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction)

Germ cell mutagenicity

The substance was determined to be non-mutagenic in bacterial as well as mammalian cell mutagenicity assays.

Carcinogenicity

Two carcinogenicity studies have been performed one in the mouse -91 weeks and another in the rat-104 weeks. Metformin was administered dietary.

There was no carcinogenic potential in either species studied.

Reproductive toxicity

Inan OECD Preliminary Reproduction Toxicity Screening Test, a NOEL of 600 mg/kg bw/day for the maternal and reproductive toxicity was determined in Sprague-Dawley rats.

In an OECD Guideline 414 (Prenatal Developmental Toxicity) Study, the test material exposure produced no evidence of any teratogenic effect in Sprague-Dawley rats. No signs of maternal toxicity, treatment related embryo lethality or fetotoxicity were evident at any dosage level studied. The NOEL of the study was 300 mg/kg bw/day for embryo and maternal toxicity.

STOT-single exposure: No data available.

STOT-repeated exposure: No data available.

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Aspiration hazard: No data available.

SECTION 12: Ecological information

12.1 Toxicity

Aquatic toxicity:

EC50 (48hr) 64 mg/L (Daphnia magna)

EC50 (72hr) >320 mg/L (Desmodesmus subspicatus)

LC50 (96hr) >982 mg/L (Lepomis macrochirus)

12.2 Persistence and degradability: The substance is not readily biodegradable

12.3 Bio accumulative potential

The test material is well soluble in water (350 g/L). Due to the hydrophilic character of the compound, little or no distribution into octanol can be observed (log K_{ow} -1.25). Based on the low log K_{ow} the test material has low potential to cross biological membranes and there exist no potential for bioaccumulation.

12.4 Mobility in soil

Koc Estimate from MCI: log K_{oc} = 1.4277

This value indicates that the substance will have low sorption to soil and sediment and moderate migration potential to ground water.

Ecotoxicological effects:

Remark: Harmful to fish

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Additional ecological information:

General notes: Harmful to aquatic organisms.

Do not allow undiluted product to reach ground water, water course or sewage system.

12.5 Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

12.6 Other adverse effects: No further relevant information available.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Recommendation

Dispose off waste material according to local, state and federal regulations.

Leave chemicals in original containers. Do not mix with other waste.

Uncleaned packaging:

Recommendation:

Dispose of as un used product.

Disposal must be made according to official regulations.

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SECTION 14: Transport information

14.1 UN-Number ADR, IMDG, IATA	Not applicable
14.2 UN proper shipping name ADR, IMDG, IATA	Not applicable
14.3 Transport hazard class(es) ADR,AND, IMDG, IATA Class	Not applicable
14.4 Packing group ADR, IMDG, IATA	Not applicable
14.5 Environmental hazards:	Not applicable
14.6 Special precautions for user	Not applicable
14.7 Transport in bulk according to Annex II of Marpol and the IBC Code	Not applicable
UN "Model Regulation":	Not applicable

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Labeling according to Regulation (EC) No1272/2008

The substance is classified and labeled according to the CLP regulation.

Hazard pictograms



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Hazard statements :

H302 Harmful if swallowed.

H412 Harmful to aquatic life with long lasting effects.

Precautionary statements

P273 Avoid release to the environment.

P264 Wash thoroughly after handling.

P270 Do not eat, drink or smoke when using this product.

P301+P312 IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell.

P330 Rinse mouth.

P501 Dispose of contents/container in accordance with local/regional/national/International regulations.

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Directive 2012/18/EU

Named dangerous substances-ANNEX I Substance is not listed.

National regulations: Other regulations, limitations and prohibitive regulations

Substances of very high concern (SVHC) according to REACH, Article 57

The substance is not listed as SVHC.

SECTION 16: Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

Department issuing MSDS: Product safety department.

Contact:

Tel: +91-02117-202033

Tel/Fax:91-2117-235333

E-Mail:sales@sohanhealthcare.com

Abbreviations and acronyms:

ADR: Accordeuropéensurle transport desmarchandises dangereusespar Route (European Agreement concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS : Globally Harmonised System of Classification and Labeling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

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CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

Acute Tox. 4: Acute toxicity, Hazard Category 4

Aquatic Chronic3: Hazardous to the aquatic environment-Chronic Hazard, Category 3

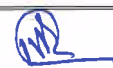
Eye Irrit.2: serious eye damage/eye irritation-category 2.

Sources

REGULATION (EC) No. 1272/2008 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on classification, labeling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No. 1907/2006.

Data from ECHA website for registered substances <https://echa.europa.eu/registration-dossier/-/registered-dossier/18705/1>

Tables 3.1 and 3.2 from annex 6 of EC 1272/2008, EC 1907/2006, EH40/2005 as amended 2011, Registry of toxic effects of chemical substances(RTECS), The dictionary of substances and their effects,1st edition, IUCLID.

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